

GES Phase 08

Enterprise CRM, Sales & Customer Experience Platform

GrounUp AI Construction Operating System

Version 1.0.0 | Review-ready | Proposed lock: GES-P08-v1.0.0

1. Executive Summary

Phase 08 establishes the governed construction customer lifecycle from omnichannel lead capture through estimating, proposals, award handoff, customer success, warranty, reviews, referrals, and repeat business.

2. Controlled Inventory

Artifact	Count
Requirements	180
Entities	72
Business rules	24
Validation rules	24
Workflows	15
Engines	20
API endpoints	24
Mapped tests	180

3. Architecture Principles

Multi-tenant, configurable, white-label, OEM/reseller capable, no-code administered, AI-native, API-first, privacy-aware, auditable, mobile-ready, and integrated with the Enterprise Data Model, Master Library Registry, Estimating, AI, and Project Operations phases.

4. Modules

Module	Code
CRM Foundation	CRM
Lead Capture & Attribution	LEAD
Qualification & Scoring	QUAL
Account & Contact 360	ACCT
Opportunity & Pipeline	PIPE
Site Visit & Discovery	DISC
Estimate Integration	EST
Proposal & E-Signature	PROP
Contract Handoff	HAND
Marketing Automation	MKT
Communication Center	COMM
Customer Portal	PORT
Customer Success	CS
Warranty & Service	WARR
Reputation & Reviews	REP
Referral Management	REF
Partner & Channel Management	CHAN

Module	Code
AI Sales Intelligence	AI
Revenue Forecasting	REV
Security & Governance	GOV

5. Customer 360 Intelligence Hub

The Customer 360 Intelligence Hub combines identity, relationships, consent, attribution, communications, activities, opportunities, estimates, proposals, projects, financial signals, surveys, reviews, warranty, referrals, AI summaries, health, lifetime value, and next actions. All insights drill to governed source records.

6. Workflows

Workflow	Definition
Omnichannel Lead Intake	Capture -> normalize -> consent check -> deduplicate -> enrich -> score -> assign -> acknowledge.
Lead Qualification	Review -> discovery -> fit assessment -> score -> qualify, nurture, disqualify, or refer.
Site Visit	Schedule -> confirm -> mobile checklist -> photos/notes -> customer acknowledgment -> follow-up.
Estimate-to-Opportunity	Create estimate request -> sync status -> receive approved scenario -> update value and scope.
Proposal Lifecycle	Generate -> internal approval -> issue -> view tracking -> negotiation -> revise -> accept/expire/decline.
Electronic Signature	Prepare -> authenticate -> sign -> evidence package -> immutable storage -> notify stakeholders.
Award-to-Project Handoff	Validate commercial terms -> collect documents -> assign operations -> create project -> close sales loop.
Marketing Nurture	Segment -> consent filter -> enroll -> send steps -> evaluate engagement -> exit/suppress/convert.
Customer Communication	Receive/send -> classify -> associate -> assign -> respond -> archive with retention.
Customer Portal Onboarding	Invite -> verify identity -> accept terms -> establish account access -> activate.
Customer Success	Create success plan -> monitor health -> schedule touchpoints -> resolve risks -> identify expansion.
Review Request	Confirm eligibility -> request review -> reminder -> ingest result -> respond/escalate.
Referral Lifecycle	Issue referral link -> capture referred lead -> validate attribution -> award -> report.
Warranty & Service	Submit -> eligibility check -> triage -> schedule -> complete -> customer acceptance -> close.
AI Next Best Action	Assemble context -> evaluate policy -> generate ranked actions -> approve -> execute -> learn.

7. Engines

Engine	Definition
Lead Capture Engine	Normalizes inbound forms, calls, emails, ads, imports, and partner submissions.

Engine	Definition
Identity Resolution Engine	Matches and merges people and organizations using governed confidence thresholds.
Attribution Engine	Calculates first-touch, last-touch, multi-touch, and configurable campaign attribution.
Lead Scoring Engine	Scores fit, intent, engagement, geography, service alignment, and risk.
Assignment Engine	Routes work by territory, trade, capacity, account rules, and service-level objectives.
Pipeline Engine	Manages stages, transitions, aging, velocity, and probability.
Opportunity Valuation Engine	Combines estimate values, alternatives, allowances, probability, and expected close date.
Proposal Generation Engine	Builds branded proposals from approved scope, pricing, options, terms, and templates.
Commercial Approval Engine	Routes discounts, margin exceptions, payment terms, and risk clauses.
E-Signature Orchestration Engine	Coordinates identity, signing order, evidence, callbacks, and tamper validation.
Marketing Automation Engine	Executes event-, schedule-, and behavior-triggered journeys.
Communication Orchestration Engine	Coordinates email, SMS, calls, portal, notifications, and suppression.
Customer Intelligence Engine	Maintains the 360-degree customer profile and timeline.
Next Best Action Engine	Ranks follow-up, nurture, cross-sell, retention, and escalation actions.
Revenue Forecast Engine	Produces weighted, category, scenario, and confidence-based forecasts.
Customer Health Engine	Scores engagement, project experience, payment, warranty, survey, and communication signals.
Reputation Engine	Manages surveys, review requests, public review ingestion, and response workflow.
Referral Engine	Attributes referrals, validates eligibility, and manages rewards.
Partner Channel Engine	Supports reseller, OEM, franchise, GC, builder, and municipal relationship workflows.
CRM Audit & Compliance Engine	Enforces retention, consent, legal hold, export controls, and audit evidence.

8. Business Rules

Rule	Definition
Lead Deduplication	Potential duplicate leads must be flagged before creation using configurable identity keys.
Source Preservation	Original lead source is immutable; later touches are recorded separately.
Consent Enforcement	Marketing communication may occur only under valid channel consent and jurisdiction rules.
Ownership	Every active lead and opportunity must have one accountable owner or governed queue.
Stage Governance	Opportunity stage changes must follow the configured transition matrix.
Lost Opportunity	A lost opportunity requires a standardized lost reason and optional competitor.
Estimate Linkage	An opportunity may link to multiple estimate scenarios but only approved revisions may populate proposal pricing.
Proposal Versioning	Issued proposals are immutable; revisions create new versions.
Approval Thresholds	Discounts, margin exceptions, and nonstandard terms require configured approval.
Signature Integrity	Signed proposal and contract artifacts require tamper-evident hashes and complete signature evidence.
Handoff Completeness	Awarded work cannot convert to a project until the handoff checklist passes.

Rule	Definition
Activity Capture	Material calls, emails, meetings, and portal events must be associated to a customer or opportunity.
Automation Safety	Automation enrollment must be idempotent and respect suppression, consent, and quiet-hour rules.
Forecast Governance	Forecast values must identify source, confidence, category, and effective date.
AI Approval	AI may not send external communication or commit commercial terms without delegated authorization.
Portal Isolation	Portal users may access only explicitly shared records within their account relationship.
Review Authenticity	Review records must preserve source and may not be altered to misrepresent customer feedback.
Referral Attribution	Referral credit must be assigned through a deterministic attribution rule.
Warranty Linkage	Warranty and service requests must link to the originating project and customer where available.
Retention	CRM records follow configurable legal, contractual, and tenant retention schedules.
White Label	Branding and outbound identity must resolve from tenant and brand profile at send time.
Channel Conflict	Partner-sourced opportunities must follow configurable account and territory conflict rules.
Customer Merge	Account or contact merges must preserve aliases, relationships, history, and audit references.
Data Export	Bulk export requires permission, purpose, watermarking where configured, and audit evidence.

9. Validation Rules

Validation	Definition
Email Format	Email fields must pass syntax validation and optional deliverability checks.
Phone Normalization	Phone numbers must normalize to E.164 when country context is known.
Required Identity	A lead requires at least one configured identity method.
Stage Date	Stage effective dates cannot precede opportunity creation.
Probability Range	Win probability must be between 0 and 100 percent.
Monetary Values	Opportunity and proposal monetary values must use tenant-supported currency and nonnegative amounts.
Proposal Expiration	Proposal expiration must be after issue date.
Signature Recipient	Signature recipients require validated identity and signing role.
Consent Date	Consent effective date cannot be after revocation date.
Quiet Hours	Automated outbound messages must pass recipient local-time quiet-hour checks.
Assignment User	Assigned owners must be active and authorized for the record scope.
Account Relationship	A contact-account relationship must not duplicate the same role and effective dates.
Referral Loop	A referral cannot refer itself or create a cyclic parent chain.
Review Rating	Review ratings must fall within the source system scale.
Warranty Eligibility	Warranty request date must be evaluated against coverage dates and exception policy.

Validation	Definition
Portal File	Portal uploads must pass file type, size, malware, and retention validation.
Template Variables	All required template variables must resolve before send or generation.
Forecast Period	Forecast snapshots require a valid fiscal period.
Duplicate Signature	A completed signature event cannot be replayed.
API Idempotency	Create and send endpoints require supported idempotency keys.
Tenant Scope	Every persistent record requires a valid tenant identifier.
Status Enumeration	Controlled status values must exist in the active configuration version.
Currency Consistency	Proposal options in one proposal version must use the same currency.
Deletion Restriction	Records under legal hold or active contract linkage cannot be permanently deleted.

10. APIs

Method	Path	Purpose
POST	/v1/leads	Create lead
GET	/v1/leads/{leadId}	Retrieve lead
POST	/v1/leads/{leadId}/qualify	Qualify or disqualify lead
POST	/v1/leads/{leadId}/assign	Assign lead
POST	/v1/accounts	Create account
GET	/v1/accounts/{accountId}/360	Retrieve customer 360
POST	/v1/contacts	Create contact
POST	/v1/opportunities	Create opportunity
PATCH	/v1/opportunities/{opportunityId}/stage	Advance opportunity stage
POST	/v1/opportunities/{opportunityId}/estimate-links	Link estimate
POST	/v1/proposals	Generate proposal
POST	/v1/proposals/{proposalId}/issue	Issue proposal
POST	/v1/proposals/{proposalId}/signature-requests	Create signature request
POST	/v1/signature-events/webhook	Receive signature event
POST	/v1/opportunities/{opportunityId}/handoff	Create project handoff
POST	/v1/automations/{automationId}/enrollments	Enroll customer
POST	/v1/communications	Send governed communication
GET	/v1/communications/threads/{threadId}	Retrieve thread
POST	/v1/portal/invitations	Invite portal user
POST	/v1/review-requests	Create review request
POST	/v1/referrals	Create referral
POST	/v1/warranty-requests	Create warranty request
GET	/v1/forecasts	Retrieve revenue forecast
GET	/v1/ai/next-best-actions	Retrieve AI recommendations

11. UI/UX Surfaces

Surface	Definition
CRM Command Center	Executive customer, pipeline, forecast, task, and risk overview.
Lead Inbox	Omnichannel intake, deduplication, scoring, routing, and SLA management.
Customer 360	Account, contacts, communications, estimates, projects, financial, warranty, and sentiment timeline.
Opportunity Workspace	Stage, activities, site visits, estimate links, competition, forecast, and approvals.
Pipeline Board	Configurable Kanban and list views with stage aging and drag controls.
Proposal Studio	Brand templates, scope, options, terms, approvals, preview, issue, and signature.
Marketing Journey Builder	No-code segments, triggers, steps, conditions, testing, and performance.
Communication Center	Unified email, SMS, calls, portal messages, templates, and consent.
Customer Portal	Proposal, signature, schedule, project updates, files, payments, warranty, and messages.
Customer Success Workspace	Health, touchpoints, risks, renewals, expansion, and referrals.
Reputation Center	Surveys, NPS, review requests, public reviews, responses, and testimonials.
CRM Administration	Fields, stages, scoring, territories, automation, consent, branding, security, and integrations.

12. Reports and KPIs

Report	Definition
Executive Revenue Dashboard	Pipeline, forecast, bookings, conversion, velocity, and customer value.
Lead Source Performance	Volume, quality, conversion, revenue, and cost by source/campaign.
Pipeline Health	Stage value, aging, stalled deals, next actions, and risk.
Sales Activity & SLA	Calls, emails, meetings, response time, follow-up compliance, and workload.
Proposal Performance	Issue-to-view, view-to-accept, time to decision, revisions, and option selection.
Win/Loss Analysis	Reasons, competitors, pricing, trade, geography, and customer segment.
Revenue Forecast	Commit, best case, pipeline, weighted, and scenario forecast.
Marketing Journey Performance	Enrollment, delivery, engagement, conversion, suppression, and attribution.
Customer Health & Retention	Health distribution, risks, touchpoints, service issues, and expansion.
Reputation & NPS	Survey response, NPS, review volume, rating, sentiment, and response SLA.
Referral Performance	Referrals, conversion, revenue, rewards, and top advocates.
Warranty & Service Experience	Request volume, eligibility, response, completion, satisfaction, and recurrence.

KPI	Definition
Lead Response Time	Median time from lead receipt to first human or governed automated response.
Lead Qualification Rate	Qualified leads divided by processed leads.
Lead-to-Opportunity Conversion	Opportunities created divided by qualified leads.
Opportunity Win Rate	Won opportunities divided by closed opportunities.
Pipeline Velocity	Qualified opportunities multiplied by win rate and average value, divided by sales-cycle days.
Stage Aging	Average days active opportunities remain in each stage.

KPI	Definition
Sales Cycle Length	Average days from qualified lead to won or lost.
Average Opportunity Value	Average expected or booked value per opportunity.
Proposal View Rate	Issued proposals viewed by at least one recipient.
Proposal Acceptance Rate	Accepted proposals divided by issued proposals.
Proposal Decision Time	Median time from issue to accept, decline, or expiration.
Forecast Accuracy	Variance between forecast bookings and actual bookings.
Marketing Qualified Lead Cost	Campaign cost divided by marketing-qualified leads.
Campaign Revenue Attribution	Attributed booked revenue by campaign.
Customer Lifetime Value	Expected gross contribution over the customer relationship.
Customer Health Score	Composite score of engagement, delivery, payment, service, and sentiment.
Net Promoter Score	Promoters percentage minus detractors percentage.
Review Conversion Rate	Published reviews divided by eligible review requests.
Referral Conversion Rate	Won referred opportunities divided by valid referrals.
Warranty First Response Time	Median time from warranty submission to first accountable response.

13. AI Governance

AI may classify, summarize, recommend, draft, score, and explain. It may not independently send external communications, alter approved pricing, accept contracts, promise schedule commitments, or determine warranty coverage without explicit delegated authorization. Every AI output includes source, confidence, model and prompt version, policy, and approval evidence.

14. Acceptance and Lock

Approval requires accepted evidence for Must requirements and all review gates. After approval, the package is locked under GES-P08-v1.0.0 and subsequent changes require governed change control.